

PAGE 1 — FROM IDEA TO INSTALLED OBJECT

One Continuous Experience

Sarah has lived in her house for eight years. Next to the fireplace is a narrow alcove where nothing fits. Too deep for standard furniture. Not square enough for shelving. The left wall bows inward by half an inch, and the baseboard cuts diagonally across the back. The floor slopes slightly where the foundation settled.

Every attempt ends the same way: measure, guess, trim, and return pieces that "almost" worked.

Nothing was wrong with the parts—they were simply correct somewhere else.

One evening she opens the interface. It asks how she wants to capture the space: enter measurements, upload plans, or scan. She uses her phone. The screen traces wall lines, ceiling angle, and floor slope. It pauses to confirm two boundaries she passed too quickly. Then it asks three alignment questions:

- Should components align with the fireplace mantle?
- Should the system compensate for the floor slope?
- Should trim match existing molding?

She selects yes. She has not drawn anything, yet the space is already defined. She selects a layout and adjusts shelf heights. The price updates instantly. Not an estimate—the price. She switches from the default pine to North Carolina cherry. The number updates—exact, because the system reconciles geometry with live local inventory, including regionally sourced material.

The intermediate translation steps that normally require separate roles are compressed into a single resolved definition. A confirmation appears: **Project Buildable**. All design elements are within defined system parameters. She presses purchase. The fabrication definition is confirmed before anything moves.

Two Hours Later The definition was transmitted to a nearby lumberyard. The operator did not interpret drawings; they loaded the boards listed on the screen. The machine referenced each piece and began execution. Holes appeared where fasteners would land in studs the system already located. Ends tapered to account for the floor slope she scanned but never calculated.

At home, she opens the bundle. The instructions contain no dimensions, only relationships: **A1 connects to A2; B3 inserts into B2**. The lowest component sits level despite the floor slope. The side panels meet the ceiling evenly. She keeps waiting for a piece that requires shaving or forcing, but it never arrives. The final piece slides in without trimming.

The project succeeded because the room was resolved before the saw began. Sarah realizes she hasn't touched a tape measure all day. The object was not adapted to the room; it arrived already adapted.

Workflow: scan → define → select → confirm → cut–mill–drill → label → pickup → install

The fabrication definition exists at the moment of confirmation before anything moves. Nothing new is designed after purchase. Two hours later, she receives a pickup message.

PAGE 2 — THE MECHANISM: WHY THE RESULT IS RELIABLE

1. Orchestration (Resolution Before Cutting) The result Sarah experienced was produced by a process fully resolved at confirmation. At that moment, the system generates three outputs simultaneously: a SKU-bound bill of materials, machine-ready toolpath instructions, and labeled assembly relationships.

This is not a drawing file; it is a **resolved fabrication definition** that reconciles spatial capture, geometric constraints, and live material availability. Because geometry and inventory are reconciled in advance, pricing is exact, and execution confirms a definition that already exists. No interpretive step remains after confirmation. Reliability is established before the first cut.

2. The Two-Patent Structure (Definition Layer + Execution Layer) The system addresses two independent failure modes: incorrect fabrication definition and dimensional drift during execution.

- **Patent 1 (U.S. 9,720,401)** protects the logical translation. It defines the method that converts intent into a complete fabrication definition tied to real inventory. It ensures relevant geometric and material constraints are resolved before anything moves.
- **Patent 2 (U.S. 10,768,609)** protects the operational execution. It ensures accuracy no longer depends on accumulated measurement. Every operation re-anchors to a shared geometric origin, preventing deviation from compounding.

The first patent establishes that the workflow can exist; the second establishes that it can be executed repeatably in a supervised environment.

3. The Bounded Habitat (Reliability Through Constraint) The system operates within two familiar shop footprints. A **dimensional processing unit** occupies the roller-path footprint of a standard radial arm saw, and a **sheet-processing unit** mirrors the footprint of a vertical panel station.

Its domain is focused on repeatable, fit-critical interior components and assemblies: shelving systems, built-ins, and foundational interior project components. By resolving the relationship between dimensional and sheet-based materials at the store level, the architecture enables a broad spectrum of fabrication while remaining within a defined safety and geometric window. If a project falls outside those tolerances, it is declined. These refusals allow an operator or steward to study the habitat's limits and refine the process under supervision.

4. Distributed Infrastructure Because each unit mirrors existing infrastructure footprints, deployment does not require facility reorganization. The hardware integrates directly into the material-handling paths retailers and institutions already maintain. This allows for placement according to regional need—housing repair corridors, contractor ecosystems, or training environments—without site retrofitting or specialized construction.

A narrow technical boundary with wide operational jurisdiction allows the system to be evaluated in three bounded contexts: as a supervised fabrication training cell within existing lab footprints; as a limited manufacturability pilot within a regional shop or lumberyard; or as a research platform for constraint-based fabrication workflows. It is offered solely for structural evaluation.